

Illegal gold mining in the Amazon: question cards

Data for Good, Lessons 3 to 5 · Five guiding questions · Read the assignment brief first

Economic framing. Small-scale gold mining is an occupational choice. People enter mining when the expected return exceeds what they can earn elsewhere. Questions 1 to 3 each look at one driver of this choice:

Driver	What changes it	Question
Gold price	The world gold price averaged \$1,269 per ounce in 2018. In August 2026 it was \$4,411.	1
Local alternatives	What miners could earn elsewhere falls in a recession, with inflation, or during a pandemic.	1
Expected cost	Law enforcement, and the legal status of the land: protected area, buffer zone, indigenous territory.	2, 3

Questions 4 and 5 are about prediction (where will mining appear next?) and monitoring (what should FZS track?).

Groups and questions. Each group has been assigned one question. Questions 1 to 3 can be answered for Peru or for Brazil; the table shows the country for your group.

Question	Group	Data you add yourselves
1. How does mining move with the gold price and with local alternatives?	Giant Otter (Brazil)	World Bank indicators by country and year
2. How does mining move with enforcement?	Giant Anteater (Peru)	Protected areas
3. Do protected areas, buffer zones and indigenous territories keep mining out?	Giant Tarantula (Peru)	Protected areas
4. Where next? A watchlist for FZS	Giant Armadillo	Amazon Mining Watch detections (108 MB)
5. What should FZS track every year?	Giant River Turtle	Current gold price; published MAAP figures

Each card has the same structure. Why FZS cares · concept · optional reading · data · three steps (describe, compare, model) · why it could mislead · check numbers · what your brief should tell FZS.

Reading. The reading listed on each card is optional. The papers are in the folder **optional_readings_by_question**, one subfolder per question; its **READ_ME_FIRST.txt** gives the links for the reports that are online only. MAAP reports are published by the NGO Amazon Conservation at maaprogram.org.

Reminders from the assignment brief. Use annual additions, not levels. Label every claim as description, prediction or causal. Say what the data cannot see (rivers, mercury).

Question 1. How does mining move with the gold price and with local alternatives?

Giant Otter: Brazil (municipalities in the Amazon biome). The card also describes the data for Peru (departments in the Amazon biome, and buffer zones).

Why FZS cares. If mining follows the world gold price, enforcement can only slow it down, and pressure will continue to rise as long as gold is expensive. If local conditions matter, then jobs and incomes are part of conservation.

Concept. The gold price is determined in world markets (central banks, investors, jewelry demand). The Amazon is too small a supplier to affect it, so the price is plausibly exogenous to any one place. But every place faces the same price in the same year, so the price cannot be separated from anything else that happened in that year: it is absorbed by year fixed effects. What you can learn comes from **differential exposure**: when the price rises, does mining grow more in places that have more gold? Local alternatives are harder to study: a recession may push people into mining, but mining income also affects the local economy (reverse causation).

Optional reading.

- Swenson et al. (2011), “Gold mining in the Peruvian Amazon: global prices, deforestation, and mercury imports”, *PLoS ONE*
- Asner et al. (2013), “Elevated rates of gold mining in the Amazon revealed through high-resolution monitoring”, *PNAS*
- Girard, Molina-Millán and Vic (2025), “Artisanal mining in Africa: green for gold?”, *Economic Journal*. Read the introduction for the research design: it compares gold-suitable and other locations in years with high and low gold prices
- Fiestas-Flores, Higa and Montoya-Zumaeta (2025), “COVID-19 impacts on Amazon deforestation”, *Trees, Forests and People*

Data. In the pack: `prices_annual.csv`; `peru_department_year.csv` and `peru_bufferzone_year.csv`, or `brazil_municipality_year.csv` (keep `biome == "Amazônia"`); `mining_area.csv` for gold-only mining in Brazil. **You add:** World Development Indicators for your country, by year, from data.worldbank.org: GDP per capita growth (NY.GDP.PCAP.KD.ZG), inflation (FP.CPI.TOTL.ZG), unemployment (SL.UEM.TOTL.ZS). Open the indicator page, select the country, download the CSV.

Step	What to do
Describe	Mining area and annual additions for your units. Plot the additions and the gold price in two panels with a common time axis. Show the same plot in levels and explain why it is misleading.
Compare	Split units by exposure: the share of the unit's area that was mining in 2010, above or below the median among units with any mining. Split years into those in which the gold price rose by more than 10%, and all others. Report the four mean additions.
Model	$y_{it} = \alpha_i + \gamma_t + \beta (\ln \text{gold}_t \times \text{exposure}_i) + \varepsilon_{it}$. Placebo: replace the gold price by the silver price, then by the copper price. Then add your measure of local alternatives, also interacted with exposure.

Why it could mislead.

- All places face the same price: only differences in exposure are informative
- Silver and gold prices move together (correlation of annual changes: 0.78 since 1985). If you find the same result with silver, you are picking up a general commodity cycle or a trend
- Exposure measured by past mining is itself the result of past prices
- Local economic conditions are affected by mining income
- Peru has few units, and Madre de Dios has 95% of the mining area in the Amazon biome. Do not expect precise estimates

Check numbers. Madre de Dios 2025: 112,622 ha · Brazil, all municipalities, 2024: 609,637 ha · gold, August 2026: \$4,411 · additions in Madre de Dios: 4,940 ha in 2021, 10,540 ha in 2022, 19,339 ha in 2023.

Your brief should tell FZS: if the gold price stays above \$4,000, where should they expect mining to increase, and how sure are you?

Question 2. How does mining move with enforcement?

Giant Anteater: Peru (Operation Mercurio, from February 2019). The card also describes Brazil (indigenous territories, 2019 to 2022, and the operation in the Yanomami territory from February 2023).

Why FZS cares. FZS supports law enforcement against illegal mining. It needs to know whether an enforcement operation reduces mining, moves it elsewhere, or only delays it.

Concept. Enforcement raises the expected cost of mining in the targeted area. Mining may fall (deterrence), move to other areas (displacement), or return when enforcement ends. In the targeted area alone, these three look the same. A before-and-after comparison with a comparison group (difference-in-differences) separates the effect of the operation from the general trend, but only if the comparison group is not affected by the operation. If miners move to the comparison area, there are spillovers. In Rwanda, spillovers to the control group would have biased the estimate toward zero. Here they bias it away from zero: the comparison area gets more mining because of the operation.

Optional reading.

- MAAP #130 (2020), “Illegal gold mining down 78% in Peruvian Amazon, but still threatens key areas”
- MAAP #193 (2023), on the return of mining to La Pampa, and MAAP #208 (2024), summary for 2021 to 2024
- Assunção, Gandour and Rocha (2023), “DETER-ing deforestation in the Amazon: environmental monitoring and law enforcement”, *AEJ: Applied Economics*
- Saavedra (2025), “Technology, information and state capacity: experimental evidence from illegal mining in Colombia”, working paper. A randomized experiment that measures spillovers

Context. *Peru.* Operation Mercurio (in English-language reports: Operation Mercury) started in February 2019 in La Pampa, the largest illegal mining area, located in the buffer zone of the Tambopata National Reserve. MAAP #130 reports that mining deforestation in La Pampa fell by 90%, and that it increased in three nearby areas. MAAP #193 reports that by 2023 miners had returned. *Brazil.* Between 2019 and 2022, federal enforcement against illegal mining is widely reported to have weakened (find and cite a source). Artisanal mining inside indigenous territories increased from 13,704 ha in 2018 to 30,980 ha in 2022. In January 2023 the federal government declared a public health emergency in the Yanomami territory. Operations to remove the miners started in February 2023.

Data. In the pack: `peru_bufferzone_year.csv` (Tambopata is the targeted buffer zone), or `brazil_municipality_year.csv` and `mining_area.csv`. **You add:** *Peru:* the file “Áreas naturales protegidas” (Colección 4) from peru.mapbiomas.org/descargas/estadisticas, to see what happened inside the reserve itself. *Brazil:* the file “Indigenous Territories” (Collection 10.1) from brasil.mapbiomas.org/en/estadisticas, for the Yanomami, Munduruku and Kayapó territories. It ends in 2024, which leaves two years after the operation.

Step	What to do
Describe	Annual additions for the targeted unit and for each possible comparison unit, from 2014 to the latest year.
Compare	Before and after, targeted and comparison. The starter code does this for the Tambopata and Amarakaeri buffer zones. Then three checks. <i>Displacement:</i> add up the targeted unit and its neighbors. Did the total fall? <i>Placebo:</i> repeat the comparison as if the operation had started three years earlier. <i>Persistence:</i> extend the comparison to the latest year.
Model	Event study: $y_{it} = \alpha_i + \gamma_t + \sum_k \beta_k (\text{targeted}_i \times 1[t = k]) + \varepsilon_{it}$, omitting the year before the operation. Plot the β_k .

Why it could mislead.

- COVID-19 started one year after Operation Mercurio
- Spillovers: additions in the Amarakaeri buffer zone increase from 54 ha in 2018 to 1,086 ha in 2022
- In 2023 the gold price was rising and enforcement had weakened. The data cannot separate the two
- Mining on an old mining site counts as an addition. MAAP #193 reports that the return to La Pampa took place on already degraded land
- With one targeted unit, standard errors are not informative. Show the figure and the placebo

Check numbers. Tambopata buffer zone 2025: 20,730 ha · mean annual additions in 2016-18 and 2019-21: Tambopata 1,640 and 163 ha; Amarakaeri 289 and 574 ha · all Brazilian indigenous territories, artisanal, 2025: 39,915 ha · Roraima, all municipalities: 446 ha in 2018, 4,745 ha in 2024.

Your brief should tell FZS: what should they expect in the targeted area, and around it, when an enforcement operation ends?

Question 3. Do protected areas, buffer zones and indigenous territories keep mining out?

Giant Tarantula: Peru. The card also describes the data for Brazil.

Why FZS cares. Protected areas are FZS’s main instrument. FZS needs to know where and when their boundaries are respected, and whether mining moves to neighboring areas.

Concept. Protected areas and indigenous territories restrict mining by law. Whether they reduce mining depends on enforcement, and on whether there would have been mining in the first place. Protected areas are not randomly placed: many were created in remote areas with little economic pressure. “Less mining inside than outside” may therefore reflect where the boundary was drawn (selection). Protection may also shift mining to just outside the boundary (leakage). Legal status differs between a protected area, its buffer zone, a native community and, in Madre de Dios, the mining corridor in which small-scale mining is permitted. Your brief should say what the legal status of each of your categories is, with a source.

Optional reading.

- Asner and Tupayachi (2017), “Accelerated losses of protected forests from gold mining in the Peruvian Amazon”, *Environmental Research Letters*
- Baragwanath and Bayi (2020), “Collective property rights reduce deforestation in the Brazilian Amazon”, *PNAS*. A regression discontinuity design at the boundaries of indigenous territories
- Blackman et al. (2017), “Titling indigenous communities protects forests in the Peruvian Amazon”, *PNAS*
- MAAP #208 (2024): of 30,846 ha of mining deforestation in the southern Peruvian Amazon between January 2021 and March 2024, 74% was inside the mining corridor. The remaining 26% (probable illegal mining) includes 15% in native communities and 8% in buffer zones of protected areas

Data. In the pack: `peru_bufferzone_year.csv` and `peru_department_year.csv`, or `mining_area.csv` and `brazil_municipality_year.csv`. **You add:** *Peru:* “Áreas naturales protegidas” (Colección 4) from peru.mapbiomas.org/descargas/estadisticas. “Comunidades nativas tituladas” is listed for Colección 4, but on 7 October only the Colección 3 file (to 2024) could be downloaded. If you use Colección 3, do not combine its numbers with Colección 4 in one series. *Brazil:* “Indigenous Territories” and “Protected Areas” (Collection 10.1, to 2024) from brasil.mapbiomas.org/en/estadisticas.

Step	What to do
Describe	Mining area and annual additions in three categories within the same department or state: inside protected areas; in buffer zones or indigenous territories; everywhere else (the total minus the first two). List the ten territories with the largest additions since 2018.
Compare	The share of new mining in each category, before and after a date that you choose and justify. The categories differ in size, so also report additions per 1,000 ha of the category’s area.
Model	$y_{it} = \alpha_i + \gamma_t + \beta (\text{category}_i \times \text{after}_t) + \varepsilon_{it}$, with territories as units.

Why it could mislead.

- Selection: areas were protected where mining was unlikely. Less mining inside does not show that protection works
- Leakage: little mining in the protected area and a lot in its buffer zone is consistent with protection and with displacement
- These tables do not show what happens at the boundary itself. Choose the best available comparison (same department, same period, similar level at the start) and justify it
- A protected area or buffer zone may extend over two departments. Add up first
- Brazil: inside indigenous territories most mining has no substance recorded. Define your measure

Check numbers. Tambopata buffer zone 2025: 20,730 ha · all buffer zones in Madre de Dios 2025: 34,635 ha, which is 31% of the department’s mining area · all Brazilian indigenous territories, artisanal, 2025: 39,915 ha · and for each territory you download, the number shown by the publisher.

Your brief should tell FZS: which boundaries have been respected, which have not, and what can the data not tell us?

Question 4. Where next? A watchlist for FZS

Giant Armadillo. There is no starter table for this question: you work with a 108 MB GeoJSON file that you download yourselves.

Why FZS cares. Patrols and overflights are costly. A ranked list of areas at risk is useful for planning them, even without knowing the causes.

Concept. This is a prediction problem, not a causal one. Mining spreads from existing sites, along rivers and roads, so mining nearby in the past is the best predictor of new mining. This is the benchmark that your model has to beat. Evaluate predictions out of sample: estimate on earlier years, test on later years. Keep base rates in mind: new mining is rare, so even a good model produces many false positives over the whole Amazon basin.

Optional reading.

- MAAP #235 (2025), “AI-powered detection of Amazon gold mining deforestation in real-time” (pre-class reading)
- The Amazon Mining Watch `README.md` at source.coop/earthgenome/amazon-mining-watch. This is the documentation of the data
- Camalan et al. (2022), “Change detection of Amazonian alluvial gold mining using deep learning and Sentinel-2 imagery”, *Remote Sensing*
- Saavedra (2025), see Question 2: what happened when predicted mine locations were given to the authorities

Data. You download: `amazon_basin_detections.geojson` from source.coop/earthgenome/amazon-mining-watch (Earth Genome and Amazon Conservation; license CC BY 4.0). For each detection the file records the model’s confidence, the period in which mining was first confirmed, and whether the detection is confirmed or provisional. Annual periods: 2018 to 2025. Quarterly periods are provisional. Start by listing the fields in the file and what each one means.

Step	What to do
Describe	Detections by country and by first period. Reduce each detection to one point and round its coordinates to 0.1 degrees (cells of about 11 km). This gives you a table of cells by year. No GIS needed.
Predict	Outcome: cells with a first detection in 2023 to 2025. Predictors, measured up to 2022: detections in the eight neighboring cells, years since the first detection in the neighborhood, country. A logistic regression is sufficient.
Evaluate	Compare your model with the simple rule “any cell next to existing mining”. Report precision for your top 20 and top 50 cells: how many of them did get new mining? Give FZS your top 20 as a table, with coordinates and country.

Why it could mislead.

- A prediction is not a causal claim. Do not interpret the coefficients as causes
- According to the README, detected areas are larger than hand-drawn boundaries. Compare periods, not totals
- Confirmed and provisional detections are defined differently
- The model detects mining on land, not dredges on rivers. This is why little mining is detected in the Colombian Amazon
- If the authorities act on a watchlist, miners will respond, and the predictions will become less accurate

Check numbers. Number of detections by country for one period, compared with the Amazon Mining Watch dashboard (amazonminingwatch.org). Report the period and the settings.

Your brief should tell FZS: if they can check twenty places next year, which twenty, and how often would you have been right in the past?

Question 5. What should FZS track every year?

Giant River Turtle. This question involves less estimation and more measurement.

Why FZS cares. FZS reports to donors and governments on whether the situation is improving or getting worse. It needs a small number of indicators that it can defend.

Concept. Every indicator depends on a definition. Three measures are easily confused: the area classified as mining in a given year (a stock; MapBiomas), the cumulative deforestation due to mining (MAAP), and new mining in a year or quarter (a flow; Amazon Mining Watch, or annual additions). Three things are not measured by any land-cover data: dredges on rivers, mercury, and whether a mine is legal. A monitoring system combines leading indicators (the gold price), coincident indicators (quarterly detections) and lagging indicators (annual maps).

Optional reading.

- MAAP #233 (2025), “Current situation of gold mining in the Peruvian Amazon”, maaprogram.org/gold-mining-peru-amazon
- Caballero Espejo et al. (2018), “Deforestation and forest degradation due to gold mining in the Peruvian Amazon: a 34-year perspective”, *Remote Sensing*
- Gerson et al. (2022), “Amazon forests capture high levels of atmospheric mercury pollution from artisanal gold mining”, *Nature Communications*

Data. In the pack: all six tables. **You add:** the latest monthly prices (“Pink Sheet”) from the World Bank commodity markets page, and the published figures that you cite from MAAP reports. Enter each in the data log, with report number, page or figure, and date.

Step	What to do
Describe	For Madre de Dios, compare the sources for the same year: MapBiomas reports 112,622 ha (2025), MAAP #233 reports 135,939 ha (as of mid-2025). Explain each difference by the definitions.
Propose	Five indicators for FZS. For each one: current value, source, how often it is updated and with what delay, a threshold, and what FZS should do when the threshold is crossed.
Check	Additions in Madre de Dios were 4,940 ha in 2021, 10,540 ha in 2022 and 19,339 ha in 2023. Using only data that were available at the time, when would each of your indicators have crossed its threshold?

Why it could mislead.

- Numbers from different sources should not be subtracted from each other
- A threshold that is chosen after looking at the data will always appear to work
- New collections revise earlier years. Report the collection for every number
- An indicator that becomes a target may be manipulated
- Rivers: MAAP #233 reports mining in 225 water bodies, and 989 dredges in Loreto since 2017. None of this appears in the tables

Check numbers. Madre de Dios 2025: 112,622 ha · Peru, Amazon biome, 2025: 118,296 ha · gold, August 2026: \$4,411.

Your brief should tell FZS: which five indicators should they look at every year, and what does each of them not show?